Christian P. Villedo

christianvilledo09@gmail.com • 09763090473
www.linkedin.com/in/christian-villedo

SUMMARY

Statistical consulting intern with experience in data analytics, predictive modeling, machine learning, and research-driven problem solving. Skilled in applying statistical and machine learning techniques to real-world datasets using R, SAS, SQL, and Excel. Experienced in data cleaning, visualization, model development, and translating complex analytical findings into actionable insights through academic research and client-facing consulting projects. Expected to graduate with a Bachelor of Science in Statistics from the University of the Philippines Diliman in July 2026.

EDUCATION

Bachelor of Science, Statistics | University of the Philippines Diliman | September 2022 - July 6, 2026

EXPERIENCE

Statistical Consultant Intern | *Altabay Solutions* | *June – July 2025*

- Conducted statistical analyses and prepared client-ready reports using RStudio and Excel.
- Designed and managed data dashboards to summarize findings and improve presentation of research results.
- Engaged directly with clients, assisting them with study design, data cleaning, and interpretation of statistical outputs.
- Gained practical experience in research consulting, emphasizing the value of accuracy, clarity, and client collaboration.

Finance Committee Member | UP Variates | 2024-2025

- Assisted in the planning and monitoring of the organization's annual budget
- Managed expense tracking and financial documentation for various events
- Collaborated with external partners and internal committees for sponsorships and financial coordination

Statistical Challenge Proctor | UP Variates | 2024-2025

- Supported the smooth implementation of UP Variates' flagship academic competition
- Ensured fairness and accuracy during examinations and scoring
- Guided participants and addressed inquiries during the event

Secretariat Committee Member | UP Variates | 2023-2024

- Handled documentation and logistical support for events and meetings
- Maintained records and coordinated communications between members and the executive committee
- Assisted in the preparation of minutes and reports

ACADEMIC PROJECTS

Multivariate Analysis of Budgetary Strategies in Philippine State Universities and Colleges | *Co-author, Stat 147: Introduction to Multivariate Analysis*

- Constructed a composite performance index for 115 State Universities and Colleges (SUCs) using Non-linear Iterative Partial Least Squares (NIPALS) to address non-linear relationships and missing data.
- Applied Principal Component Analysis (PCA) to condense 9 highly correlated expenditure variables into 5 core budgetary strategies, reducing dataset dimensionality while retaining 77% of the total variance.
- Performed Cluster Analysis utilizing Agglomerative Nesting (AGNES) with Ward's linkage and K-means clustering to classify SUCs into 7 distinct financial typologies based on their allocation patterns.
- Evaluated the impact of budget strategies on institutional outcomes using Principal Component Regression and non-parametric Kruskal-Wallis tests via R.

Causal Analysis of Musical Valence on Spotify Streaming Success | *Co-author, Stat 197: Causal Inference*

- Investigated the causal effect of a song's emotional valence on Spotify streaming counts using observational data from Taylor Swift's discography.
- Extracted and pre-processed track-level data using the Spotify API (spotifyr package) and performed web scraping on Kworb.net to compile comprehensive stream counts.
- Implemented Propensity Score Matching (PSM) using Nearest-Neighbor and Full Matching techniques via logistic and probit regression models to control for confounding audio features like danceability and energy.
- Assessed covariate balance and estimated the Average Treatment Effect on the Treated (ATT) utilizing R packages such as MatchIt and cobalt.

Assessment of Models for Heat Index of Philippine Barangays | *Co-author, Stat 142: Introduction to Computational Statistics*

- Modeled barangay-level heat indices using the Project CCHAIN dataset across 879 barangays, analyzing environmental, socioeconomic, and air quality predictors.
- Employed k-fold cross-validation to systematically evaluate variable transformations and subsetting, identifying the optimal model with the lowest Mean Square Error (MSE).
- Utilized bootstrap estimation techniques (paired bootstrap hypothesis testing and empirical bias-corrected confidence intervals) to perform robust coefficient analysis and validate model predictions despite classical regression assumption violations like heteroskedasticity.

Econometric Analysis of Determinants of LGU Total Expenditure | *Co-author, Stat 136: Introduction to Regression Analysis*

- Modeled the total expenditures of provincial-level Local Government Units (LGUs) for FY 2022 based on political and financial variables using Ordinary Least Squares (OLS) multiple linear regression.
- Executed backward elimination variable selection and applied log-transformations to resolve assumption violations, ultimately producing a robust model explaining 72.45% of expenditure variance.
- Conducted rigorous diagnostic checking in R to ensure model validity, performing tests for multicollinearity (VIF), heteroscedasticity (Breusch-Pagan), autocorrelation (Durbin-Watson), and influential outliers (Cook's Distance, DFFITS, DFBETAS).

SKILLS

R, Statistical Analysis Software (SAS), and SQL:
Data Cleaning, Data Manipulation, Data Management, Data Analysis, Hypothesis Testing, Data Visualization, Machine Learning (Supervised and Unsupervised Learning), Predictive Modeling, Model Validation (Cross-Validation), Multivariate Analysis (PCA, Factor Analysis, Clustering Techniques), Time Series Analysis (Introductory Level), Computational Statistics (Bootstrapping)

Microsoft Excel:
Data Management, Data Visualization, Dashboard Creation, Data Analysis (Descriptive Statistics, ANOVA, Correlation Analysis, Regression Analysis, t-tests, and z-tests)

Technical Writing:
Research Proposal Writing, Statistical Reporting, Questionnaire Construction and Validation, Item Analysis, and Reliability Testing

Soft Skills:
Communication, Team Collaboration, Client Handling, Organizational Management, Problem Solving